

Chapter 1

Java Introduction

Introduction to Programming Languages

A programming language is used to give instructions to a computer.

It acts as a bridge between humans and computers.

Programs written in human-readable languages are translated into machine code by a compiler or interpreter.

Types

1. Low-Level Languages
2. Close to machine language.
3. Faster but difficult for humans.

Examples:

- Machine Language (Binary)
- Assembly Language
- High-Level Languages
- Easy to read and write.
- Independent of hardware.

Examples:

- Java
- Python
- C++
- JavaScript
- Ruby

What is Java?

Java is a high-level, object-oriented, class-based programming language.

Developed by Sun Microsystems in 1995 (later acquired by Oracle).

Platform-independent because of the Java Virtual Machine (JVM).

Used for:

- Desktop applications
- Web applications
- Android apps

- Enterprise software

History of Java

1991: Project started as Oak by James Gosling.

1995: Renamed to Java.

1996: JDK 1.0 released.

1999: Java 2 introduced (J2SE, J2EE, J2ME).

2006: Java became open source.

2010: Oracle acquired Sun Microsystems.

2014 onwards: New Java versions released every 6 months.

Why Learn Java?

Widely used in software development.

Platform independent (Write Once, Run Anywhere – WORA).

Excellent career opportunities.

Fully supports Object-Oriented Programming (OOP).

Secure, reliable, and stable.

Huge community and many frameworks (like Spring and Hibernate).

Continuously updated and future-proof.